High Impedance Differential Probe
For Oscilloscopes And Multimeters

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The most commonly used multimeters and oscilloscopes are sold with simple non-symmetrical (single-ended) probes. But often we need differential probes to measure and observe signals between non-grounded test points.

There are different types of differential probes available, but most of them are costly. Also, some of these get damaged easily. And repair of probes (when possible) can be very expensive. Here is a simple and low-cost solution for the differential probes.

Circuit and working

The circuit diagram of a high impedance differential probe for oscilloscopes and multimeters is shown in Fig. 1. The probe has a high input resistance of 10-mega-ohm.

It is built around two popular integrated circuits—IC1 and IC2. IC1 is a dual operational amplifier (op-amp), which can be TL072, TL082, TL062, or a similar one. It should have JFET inputs and low noise.

IC2 is a single op-amp such as LF356 and LF357. It should preferably have pins for offset control. IC2 can be TL071, TL061, TL081, etc. But the offset control should be changed.

The offset control is needed only in DC mode. IC1 and IC2 should have a low-temperature drift.

In order to keep the circuit as simple as possible, both inbuilt operational amplifiers in IC1 work as simple followers. The circuit has DC or AC inputs. If you close switches S1 and S2, the input of the probes is DC. If you open S1 and S2, the input of the probes is AC. Switches S1 and S2 should open and close simultaneously.

The inputs of IC1 are protected by resistors (R1 and R2), diodes (D1 through D4), and Zener diodes (ZD1

PARTS LIST

Semiconductors:	
IC1	- TL072 op-amp
IC2	- LF356 op-amp
D1-D6	- 1N4148 signal diode
ZD1, ZD2	- 10V Zener diode
Resistors (all 1/4-watt, ±5% carbon):	
R1, R2, R7	- 1-kilo-ohm
R3, R4	- 2-kilo-ohm
R5, R6	- 20-kilo-ohm
R8	- 100-ohm
R9, R10	- 10-mega-ohm
VR1	- 27-kilo-ohm potmeter
Capacitors:	
C1, C2	- 10nF ceramic disk
C3, C4, C6, C7	- 100nF ceramic disk
C5, C6	- 100µF, 40V electrolytic
Miscellaneous:	
CON1, CON2	- 3-pin connector
CON3	- 2-pin connector
S1, S2	- On/off switch
	- 8-pin IC base (2 number)
	- Jumper wires

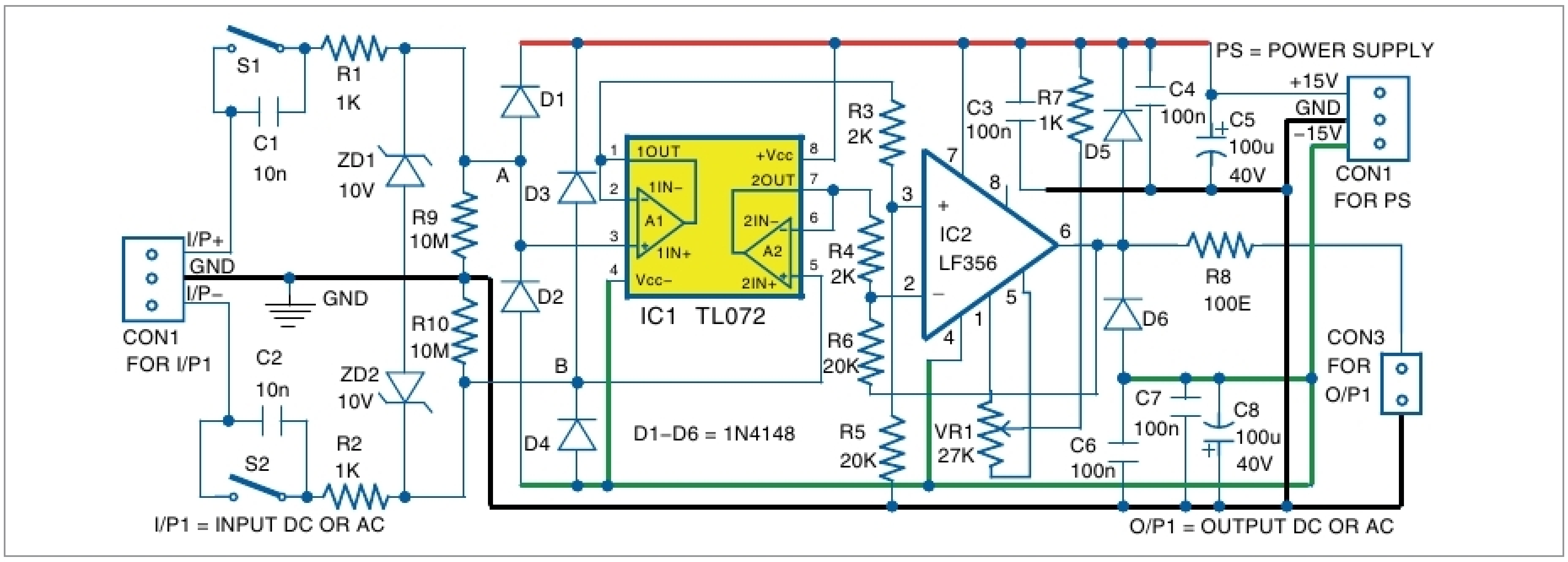

Fig. 1: Circuit diagram of a high impedance differential probe for oscilloscope and multimeter